Haar -> Transformada discreta de Haar

Partiendo de una matriz A de 8x8 de muestra.

$$\boldsymbol{A} = \begin{pmatrix} 200.0 & 200.0 & 200.0 & 200.0 & 50.0 & 50.0 & 50.0 & 50.0 \\ 200.0 & 200.0 & 200.0 & 200.0 & 50.0 & 50.0 & 50.0 & 50.0 \\ 200.0 & 200.0 & 200.0 & 200.0 & 50.0 & 50.0 & 50.0 & 50.0 \\ 200.0 & 200.0 & 200.0 & 200.0 & 50.0 & 50.0 & 50.0 & 50.0 \\ 100.0 & 100.0 & 100.0 & 100.0 & 20.0 & 20.0 & 20.0 & 20.0 \\ 100.0 & 100.0 & 100.0 & 100.0 & 20.0 & 20.0 & 20.0 & 20.0 \\ 100.0 & 100.0 & 100.0 & 100.0 & 20.0 & 20.0 & 20.0 & 20.0 \\ 100.0 & 100.0 & 100.0 & 100.0 & 20.0 & 20.0 & 20.0 & 20.0 \end{pmatrix}$$

Condición de ortonormalidad de la matriz de transformación:

$$H \, . H^T = I$$

Matriz H8 en forma analítica:

$$\boldsymbol{H_8} = \begin{pmatrix} \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} \\ \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & -\frac{1}{\sqrt{8}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

Matriz H8 expresada numéricamente (redondeada a 5 decimales):

$$\boldsymbol{H_8} = \begin{pmatrix} 0.35355 & 0.35355 & 0.35355 & 0.35355 & 0.35355 & 0.35355 & 0.35355 & 0.35355 \\ 0.35355 & 0.35355 & 0.35355 & 0.35355 & -0.35355 & -0.35355 & -0.35355 & -0.35355 \\ 0.50000 & 0.50000 & -0.50000 & -0.50000 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.50000 & 0.50000 & -0.50000 & -0.50000 \\ 0.70711 & -0.70711 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.70711 & -0.70711 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.70711 & -0.70711 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.70711 & -0.70711 \end{pmatrix}$$

Verificamos ortogonalidad y ortonormalidad por filas de H8:

$$\sum_{n=0}^{7}(h_0,n)^2 = \sum_{n=0}^{7}(\frac{1}{\sqrt{8}})^2 = 8\,.\frac{1}{8} = 1$$

$$\sum_{n=0}^{7} h_{0,n}\,.\; h_{1,n} = 4(\frac{1}{\sqrt{8}}\,.\;\frac{1}{\sqrt{8}}) + 4\left(\frac{1}{\sqrt{8}}\,.\left(-\frac{1}{\sqrt{8}}\right)\right) = \frac{4}{8} - \frac{4}{8} = 0$$

Queremos transformar la imagen original en una representación basada en wavelets. Para eso calculamos C:

$$C = H_8.A.H_8^T$$

Multiplicando por partes, primero calculamos H8 · A:

$$\boldsymbol{H_8} = \begin{pmatrix} 424.26407 & 424.26407 & 424.26407 & 424.26407 & 98.99495 & 98.99495 & 98.99495 & 98.99495 \\ 141.42136 & 141.42136 & 141.42136 & 141.42136 & 42.42641 & 42.42641 & 42.42641 & 42.42641 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \end{pmatrix}$$

Multiplicando ahora por H8^T para obtener C = $(\boldsymbol{H_8}\,.A)\,.H_8^T$:

$$\boldsymbol{C} = \begin{pmatrix} 739.98581 & 459.61941 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 259.91206 & 139.95714 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \end{pmatrix}$$

Elegimos el 25% de los coeficientes totales de la matriz C de coeficientes: igual a 16 en valor absoluto.

$$|\boldsymbol{C}| = \begin{pmatrix} 739.98581 & 459.61941 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 259.91206 & 139.95714 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \end{pmatrix}$$

$$m_{ij} = \begin{cases} c_{ij}\,, & si\ c_{ij}\ est\acute{a}\ entre\ los\ 16\ seleccionados \\ 0, & en\ caso\ contrario \end{cases}$$

$$\boldsymbol{M} = \begin{pmatrix} 739.98581 & 459.61941 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 259.91206 & 139.95714 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \end{pmatrix}$$

Reconstrucción inversa de la imagen:

$$\tilde{A} = H_8^T \,.\, M \,.\, H_8$$

Como H8 es ortonormal, su inversa es igual a su transpuesta: $H_8^{-1} = H_8^T$

Primero calculamos $E = H_8^T . M$

$$\boldsymbol{E} = \begin{pmatrix} 0.35355 & 0.35355 & \cdots & 0.00000 \\ 0.35355 & 0.35355 & \cdots & 0.00000 \\ \vdots & \ddots & \vdots & \vdots \\ 0.35355 & -0.35355 & \cdots & -0.70711 \end{pmatrix} x \begin{pmatrix} 739.98581 & 459.61941 & \cdots & 0.0 \\ 259.91206 & 139.95714 & \cdots & 0.0 \\ \vdots & \ddots & \vdots & \\ 0.0 & 0.0 & \ldots & 0.0 \end{pmatrix}$$

$$\boldsymbol{E} = \begin{pmatrix} 353.53553 & 212.00000 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 353.53553 & 212.00000 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 353.53553 & 212.00000 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 353.53553 & 212.00000 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 169.72141 & 100.00000 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 169.72141 & 100.00000 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 169.72141 & 100.00000 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 169.72141 & 100.00000 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \end{pmatrix}$$

Completando el product:

$$\widetilde{\boldsymbol{A}} = \begin{pmatrix} 353.53553 & 212.00000 & \cdots & 0.00000 \\ \vdots & \ddots & \vdots & \vdots \\ 169.72141 & 100.00000 & \cdots & 0.00000 \end{pmatrix} x \begin{pmatrix} 0.35355 & 0.35355 & \cdots & 0.35355 \\ 0.35355 & 0.35355 & \cdots & -0.35355 \\ \vdots & \ddots & \vdots & \\ 0.0 & 0.0 & \ldots & -0.70711 \end{pmatrix}$$

$$\widetilde{\boldsymbol{A}} = \begin{pmatrix} 200.0 & 200.0 & 200.0 & 200.0 & 50.0 & 50.0 & 50.0 & 50.0 \\ 200.0 & 200.0 & 200.0 & 200.0 & 50.0 & 50.0 & 50.0 & 50.0 \\ 200.0 & 200.0 & 200.0 & 200.0 & 50.0 & 50.0 & 50.0 & 50.0 \\ 200.0 & 200.0 & 200.0 & 200.0 & 50.0 & 50.0 & 50.0 & 50.0 \\ 100.0 & 100.0 & 100.0 & 100.0 & 20.0 & 20.0 & 20.0 & 20.0 \\ 100.0 & 100.0 & 100.0 & 100.0 & 20.0 & 20.0 & 20.0 & 20.0 \\ 100.0 & 100.0 & 100.0 & 100.0 & 20.0 & 20.0 & 20.0 & 20.0 \\ 100.0 & 100.0 & 100.0 & 100.0 & 20.0 & 20.0 & 20.0 & 20.0 \end{pmatrix}$$

Calculamos la matriz de los errores W = A - $\tilde{A}$:

$$\boldsymbol{W} = \begin{pmatrix} 200 & 200 & \cdots & 50 \\ \vdots & \ddots & \vdots & \vdots \\ 100 & 100 & \cdots & 20 \end{pmatrix} - \begin{pmatrix} 200 & 200 & \cdots & 50.0 \\ \vdots & \ddots & \vdots & \\ 100 & 100 & \ldots & 20.0 \end{pmatrix} = \begin{pmatrix} 0.0 & 0.0 & \cdots & 0.0 \\ 0.0 & 0.0 & \ldots & 0.0 \\ \vdots & \ddots & \vdots & \vdots \\ 0.0 & 0.0 & \cdots & 0.0 \end{pmatrix}$$

Como la imagen es de 8x8 y tiene 64 píxeles:

$$MSE = \frac{1}{64} \sum_{i=1}^{8} \sum_{j=1}^{8} w_{ij}^2 = \frac{0.0}{64} = 0.00000$$

Usamos PSNR para calcular los decibelios:

$$PSNR = 10 \,.\log_{10}\left(\frac{255^2}{MSE}\right) = 10.\log_{10}\left(\frac{65025}{MSE}\right)$$